Math 118: Mathematical Methods of Data Theory

Lecture 16: Constrained Optimization

Material taken from Chong & Zak and

www.math.ucla.edu/~wotaoyin/math273

Instructor: Daniel Mckenzie

Dept. of Mathematics, UCLA

February 22, 2020

Overview

- We consider the following constrained optimization problem:

$$\begin{aligned} &\text{minimize } f(\boldsymbol{x}) \\ &\text{subject to: } \boldsymbol{h}(\boldsymbol{x}) = \mathbf{0} \\ &\qquad\qquad\quad \boldsymbol{g}(\boldsymbol{x}) \leq \mathbf{0} \end{aligned}$$

  where $f : \mathbb{R}^n \to \mathbb{R}$, $\boldsymbol{h} : \mathbb{R}^n \to \mathbb{R}^m$ and $\boldsymbol{g} : \mathbb{R}^n \to \mathbb{R}^p$. Note that:

$$\boldsymbol{h}(\boldsymbol{x}) = \begin{bmatrix} h_1(\boldsymbol{x}) \\ \vdots \\ h_m(\boldsymbol{x}) \end{bmatrix} \quad \boldsymbol{g}(\boldsymbol{x}) = \begin{bmatrix} g_1(\boldsymbol{x}) \\ \vdots \\ g_p(\boldsymbol{x}) \end{bmatrix}$$

- We assume that $f, \boldsymbol{h}$ and $\boldsymbol{g}$ are continuously differentiable.
- **Goals:**
  - First order optimality conditions (*i.e.* first derivative test).
  - Second order optimality conditions (*i.e.* second derivative test).
- Any $\boldsymbol{x}$ is called **feasible** if it satisfies the constraints.

Example

- Linear Programs:

$$\text{minimize } f(\boldsymbol{x}) = c^{\top}(\boldsymbol{x})$$

$$\text{subject to: } A\boldsymbol{x} = \boldsymbol{b} \Rightarrow \boldsymbol{h}(\boldsymbol{x}) = A\boldsymbol{x} - \boldsymbol{b} = \mathbf{0}$$

$$\boldsymbol{g}(\boldsymbol{x}) = -\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \leq \mathbf{0}$$

Active Constraints and Regular Points

Let $\boldsymbol{x}$ be a feasible point:

- Recall that $\boldsymbol{g}(\boldsymbol{x}) \leq \boldsymbol{0}$ gives the *inequality constraints*.
- For any $\boldsymbol{x}$, we say $j \in \{1, \ldots, p\}$ is **active** if $g_j(\boldsymbol{x}) = 0$. Otherwise **inactive**
- Define $J(\boldsymbol{x}) := \{j : g_j(\boldsymbol{x}) = 0\}$ set of active (inequality) constraints.
- $\boldsymbol{x}$ is **regular** if $\{\nabla g_j(\boldsymbol{x}) : j \in J(\boldsymbol{x})\} \cup \{\nabla h_i(\boldsymbol{x}) : 1 \leq i \leq m\} \subset \mathbb{R}^n$ is lin. ind.
- **Notation:** $D\boldsymbol{g} = \begin{bmatrix} \nabla g_1(\boldsymbol{x}) & \ldots & \nabla g_p(\boldsymbol{x}) \end{bmatrix} \in \mathbb{R}^{p \times n}$ and $D\boldsymbol{h} = \begin{bmatrix} \nabla h_1(\boldsymbol{x}) & \ldots & \nabla h_m(\boldsymbol{x}) \end{bmatrix}$.

For feasible and infeasible points, define the **Lagrangian**:

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\mu}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + \boldsymbol{\lambda}\boldsymbol{g}(\boldsymbol{x}) + \boldsymbol{\mu}\boldsymbol{h}(\boldsymbol{x})$$

Equality Constrained Optimization

$$\text{minimize } f(\boldsymbol{x})$$
$$\text{subject to: } \boldsymbol{h}(\boldsymbol{x}) = \mathbf{0}$$

- Here, $\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^\top \boldsymbol{h}(\boldsymbol{x})$.
- $\boldsymbol{x}^*$ is a local extrema of constrained problem $\Leftrightarrow$ Exists $\boldsymbol{\lambda}^* \in \mathbb{R}^m$ s.t. $\nabla_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = 0$ **and** $\nabla_{\boldsymbol{\lambda}} \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = 0$.

$$\Leftrightarrow \nabla f(\boldsymbol{x}^*) + (\boldsymbol{\lambda}^*)^\top \nabla \boldsymbol{h}(\boldsymbol{x}^*) = 0 \quad \textbf{and } \boldsymbol{h}(\boldsymbol{x}^*) = \mathbf{0}$$

- These are **First Order Condition**. **Second Order Conditions** involve second derivatives and distinguish the type of extrema.

Karush-Kuhn-Tucker (KKT) conditions

Theorem (First Order Necessary Conditions, or FONC)

Let $f, \boldsymbol{h}, \boldsymbol{g}$ be continuously differentiable and $\boldsymbol{x}^*$ be a regular local minimizer for:

$$\text{minimize } f(\boldsymbol{x}) \quad \textit{subject to: } \boldsymbol{h}(\boldsymbol{x}) = \boldsymbol{0}, \boldsymbol{g}(\boldsymbol{x}) \leq \boldsymbol{0}$$

Then, there exists $\boldsymbol{\lambda}^* \in \mathbb{R}^n$ and $\boldsymbol{\mu}^* \in \mathbb{R}^p$ such that:

1. $\nabla f(\boldsymbol{x}^*) + D\boldsymbol{h}(\boldsymbol{x}^*)^\top \boldsymbol{\lambda}^* + D\boldsymbol{g}(\boldsymbol{x}^*)^\top \boldsymbol{\mu}^* = \boldsymbol{0}$
2. $\boldsymbol{\mu}^* \geq 0$
3. $\boldsymbol{g}(\boldsymbol{x}^*)^\top \boldsymbol{\mu}^* = 0$
4. $\boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0}$
5. $\boldsymbol{g}(\boldsymbol{x}^*) \leq \boldsymbol{0}$

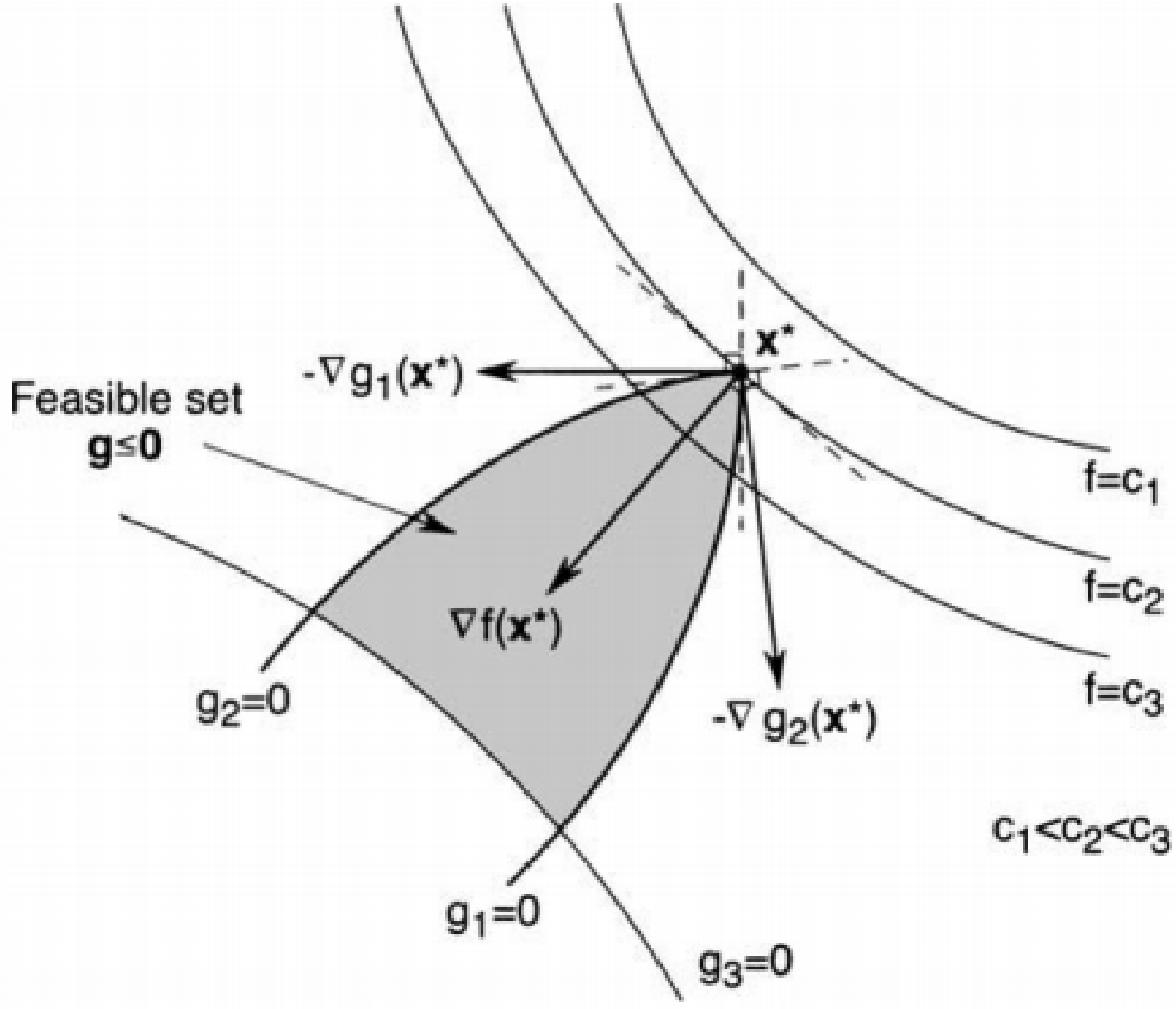
- $\boldsymbol{\lambda}^*$ are the Lagrange multipliers.
- $\boldsymbol{\mu}^*$ are called the KKT multipliers.
- Note that since $\mu_j^* \geq 0$ and $g_j(\boldsymbol{x}^*) \leq 0$ for all j, the condition:

$$\boldsymbol{g}(\boldsymbol{x}^*)^\top \boldsymbol{\mu}^* = \mu_1^* g_1(\boldsymbol{x}^*) + \ldots + \mu_p^* g_p(\boldsymbol{x}^*) = 0$$

  implies:

  - If $g_j(\boldsymbol{x}^*) < 0$ (inactive), then $\mu_j^* = 0$
  - If $\mu_j^* > 0$ then $g_j(\boldsymbol{x}^*) = 0$ (active).
  - possible: $\mu_j^* = 0$ and $g_j(\boldsymbol{x}^*) = 0$
  - impossible: $\mu_j^* > 0$ and $g_j(\boldsymbol{x}^*) < 0$

Illustration



Second Order Conditions

- Let $\boldsymbol{F}$, $\boldsymbol{H}_i$ and $\boldsymbol{G}_j$ be **Hessians** of f, h_i and g_j respectively ($1 \leq im$ and $1 \leq j \leq p$).
- Lagrangian: $\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\boldsymbol{x}) + \boldsymbol{h}(\boldsymbol{x})^\top \boldsymbol{\lambda} + \boldsymbol{g}(\boldsymbol{x})^\top \boldsymbol{\mu}$
- Hessian of $\mathcal{L}$: $\boldsymbol{L}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \boldsymbol{F}(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i \boldsymbol{H}_i(\boldsymbol{x}) + \sum_{j=1}^{p} \mu_j \boldsymbol{G}_j(\boldsymbol{x})$
- **Tangent Space:**

$$T(\boldsymbol{x}^*) := \left\{ \boldsymbol{y} \in \mathbb{R}^n : \ D\boldsymbol{h}(\boldsymbol{x}^*)\boldsymbol{y} = \boldsymbol{0}, \nabla g_j(\boldsymbol{x}^*)^\top \boldsymbol{y} = 0 \ \forall j \in J(\boldsymbol{x}^*) \right\}$$

Recall $J(\boldsymbol{x}^*) = \{j : \ g_j(\boldsymbol{x}^*) = 0\}$ *is active constraints*

Theorem (Second Order Necessary Conditions)

Let $\boldsymbol{x}^*$ *be a regular local minimizer for:*

$$\text{minimize } f(\boldsymbol{x}) \text{ subject to: } \boldsymbol{h}(\boldsymbol{x}) = \boldsymbol{0}, \ \boldsymbol{g}(\boldsymbol{x}) \leq \boldsymbol{0}$$

Then, there exist $\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*$ *satisfying FONC* **and***:*

- *For all* $\boldsymbol{y} \in T(\boldsymbol{x}^*)$ *we have* $\boldsymbol{y}^\top L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)\boldsymbol{y} \geq 0$

- set of **strongly active inequality constraints:**

$$\tilde{J}(\boldsymbol{x}, \boldsymbol{\mu}) := \{j : g_j(\boldsymbol{x}) = 0 \text{ and } \mu_j > 0\} \subset J(\boldsymbol{x})$$

- corresponding tangent space:

$$\tilde{T}(\boldsymbol{x}^*, \boldsymbol{\mu}^*) := \left\{\boldsymbol{y} \in \mathbb{R}^n : \ D\boldsymbol{h}(\boldsymbol{x}^*)\boldsymbol{y} = \boldsymbol{0}, \ \nabla g_j(\boldsymbol{x}^*)^\top \boldsymbol{y} = 0 \ \forall j \in \tilde{J}(\boldsymbol{x}^*, \boldsymbol{\mu}^*)\right\}$$

Theorem (Second Order Sufficient Conditions)

Suppose that $f, \boldsymbol{g}, \boldsymbol{h}$ are twice continuously differentiable and there exists $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) \in \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p$ such that:

1. *$\nabla f(\boldsymbol{x}^* + D\boldsymbol{h}(\boldsymbol{x}^*)^\top \boldsymbol{\lambda}^* + D\boldsymbol{g}(\boldsymbol{x}^*)^\top \boldsymbol{\mu}^* = \boldsymbol{0}$, $\boldsymbol{\mu}^* \geq 0$ and $\boldsymbol{g}(\boldsymbol{x}^*)^\top \boldsymbol{\mu}^* = 0$ (FONC);*
2. *For all $\boldsymbol{y} \in \tilde{T}(\boldsymbol{x}^*, \boldsymbol{\mu}^*)$, $\boldsymbol{y} \neq \boldsymbol{0}$ we have $\boldsymbol{y}^\top \boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)\boldsymbol{y} > 0$.*

Then $\boldsymbol{x}^*$ is a strict local minimizer.